

COMP 3411/9414

Artificial Intelligence

AI in Computer Vision

Outline

- Introduction
 - Applications of AI in computer vision
- Overall framework
 - Handwritten digit recognition as a case study
- Advanced topics
 - Texture classification
 - AI in biomedical imaging
- Reference readings

Introduction

- AI has many applications:
 - e.g. medicine, robotics, finance, transportation, marketing, arts
- Many new trends of AI technologies are developed in Computer Vision applications:
 - e.g. face recognition, surveillance, remote sensing, medical imaging, self-driving cars
 - A highly active area in both industry and research

Introduction

- Application – Face recognition
 - One of the most well-studied areas in computer vision
 - Popular algorithms: eigenface, distance metric learning
 - Industrial applications in public security, police operation, access control, advertising



Source: O. Parkhi et al. *Deep face recognition*. BMVC, 2015.

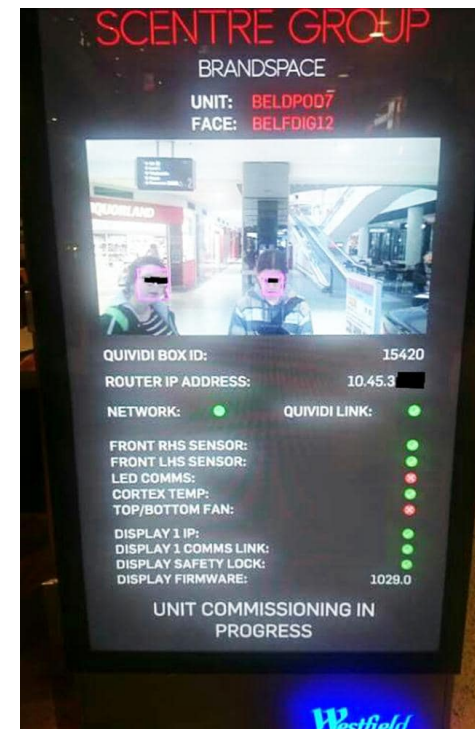
Introduction

- Application – Surveillance
 - Many different applications: object tracking, event detection, summarisation, human behaviour analysis, vehicle re-identification
 - A vast range of image/video analysis methods have been developed.



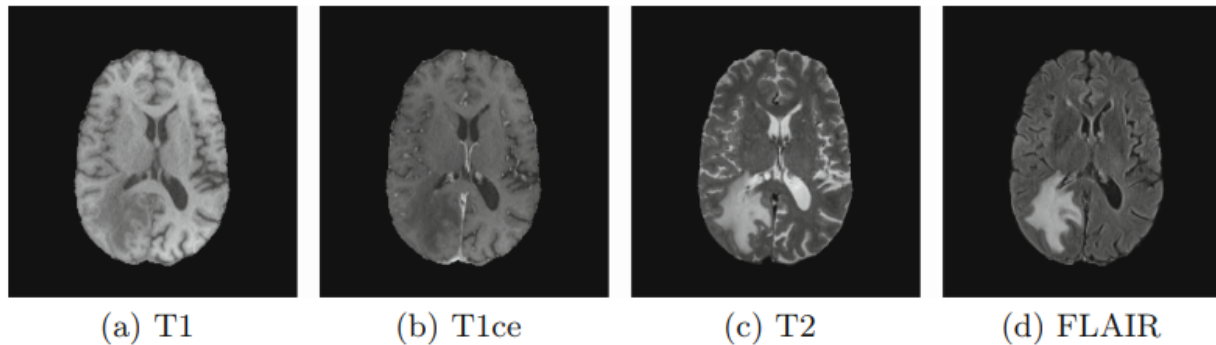
Source: A. Mabrouk and E. Zagrouba. *Abnormal behaviour recognition for intelligence video surveillance systems: a review*. Expert Systems with Applications, 2018.

<https://www.news.com.au/finance/business/retail/westfield-is-using-facial-detection-software-to-watch-how-you-shop/news-story/7d0653eb21fe1b07be51d508bfe46262>



Introduction

- Application – Medical imaging
 - Computer vision/machine learning techniques are becoming increasingly important to medical diagnosis and treatment planning.
 - Applications in lesion segmentation, tumour classification, survival prediction, drug development, nanotechnology



Source: D. Liu et al. *3D large kernel anisotropic network for brain tumor segmentation*. ICONIP, 2018.

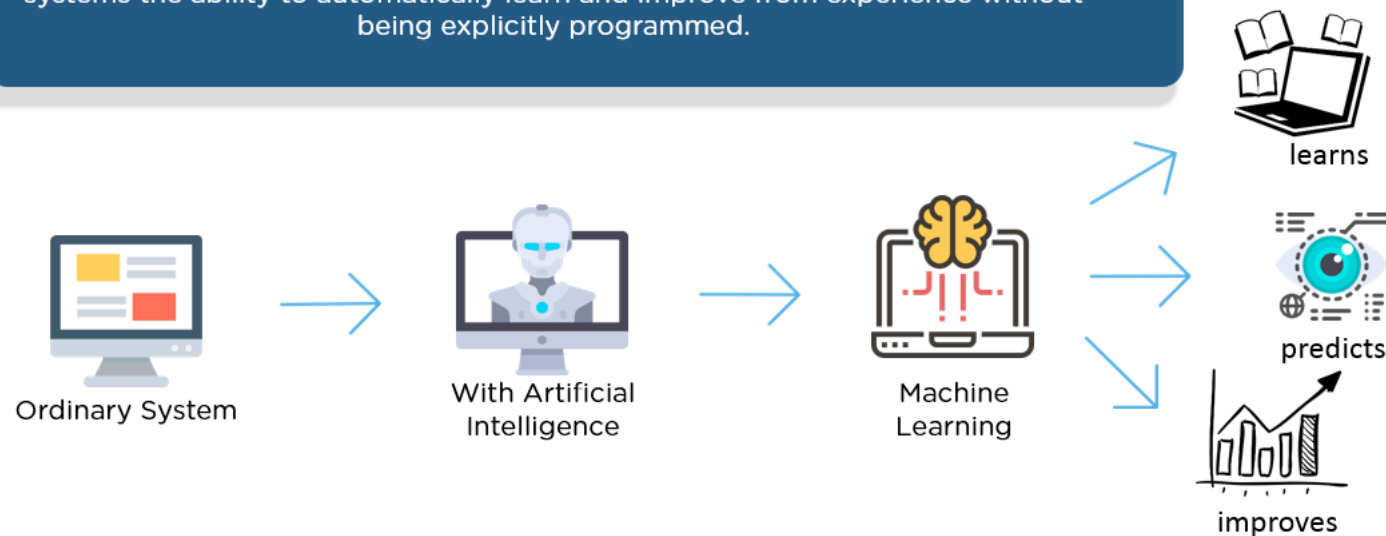
Key Learning Outcomes

- Main components of a computer vision framework
 - **Machine learning** – learning-based classification, designing a machine learning algorithm to classify the input data
 - Feature extraction – represent the images by feature vectors
 - Evaluation – ways to evaluate the performance of the system

Machine Learning

- Machine Learning, as a subfield of AI, is one of the main components in Computer Vision studies.

Machine Learning is an application of Artificial intelligence (AI) that provides systems the ability to automatically learn and improve from experience without being explicitly programmed.



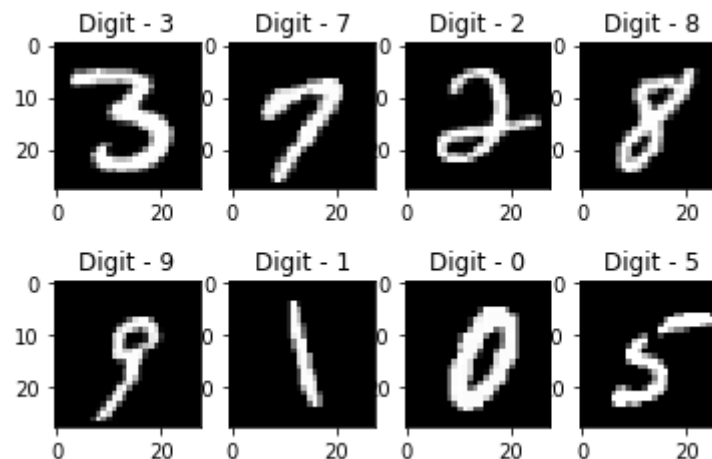
Source: <https://www.quora.com/How-is-machine-learning-related-with-artificial-intelligence>

Machine Learning

- Use **handwritten digit recognition** as a case study
 - A classical machine learning problem
 - Can be solved without the feature extraction component
- Key points:
 - Typical way of implementing a machine learning method
 - Overview of popular classifier models
 - Resources for further study

Handwritten Digit Recognition

- Objective of study:
 - Recognising handwritten digits 0-9 automatically
 - Example: <https://neuralnetwork--ieatpython.repl.co/>
 - Applications in the industry: read postcode, bank cheques



Handwritten Digit Recognition

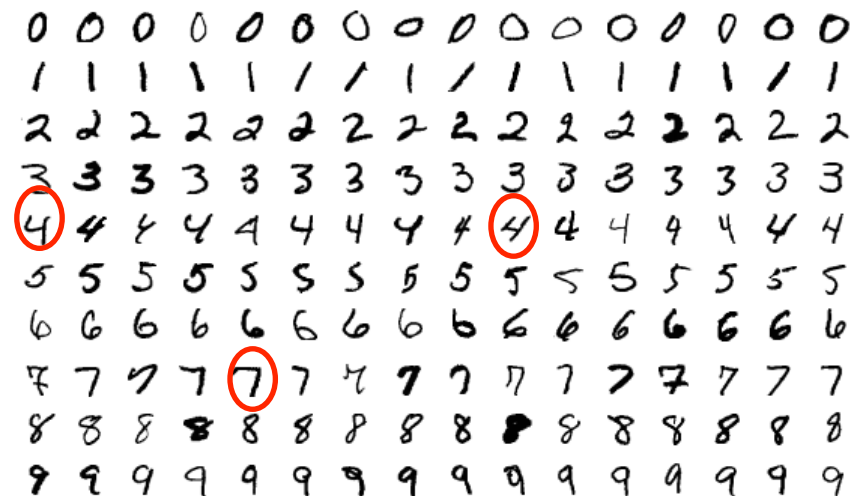
- An intuitive solution: K-Nearest Neighbour (kNN)

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1. Find K samples that are *most similar** to the test image
2. Determine which class the majority of these K samples belong to
3. Assign the test image to this class

* *Similarity is measured by certain distance functions between vectors, e.g. Euclidean distance $\|x_1 - x_2\|$*



If using the image pixels as input directly with Euclidean distance, $K=3$ nearest neighbours give a 5% error rate.

Handwritten Digit Recognition

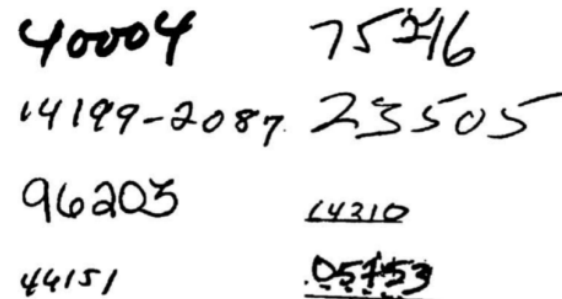
- Methodology:
 - Step 1: Construct a database for training
 - Step 2: Image pre-processing
 - Step 3: Create a machine learning / classifier model
 - Step 4: Apply the machine learning on a test image

Handwritten Digit Recognition

- Handwritten digit recognition with a back-propagation network (Y. Le Cun et al. 1990)

Step 1 – Create dataset

- Construct a dataset of 9298 segmented numerals digitised from handwritten postcodes and 3349 printed digits of 35 different fonts
- The training set contains 7291 handwritten digits and 2549 printed digits.
- The test set contains 2007 handwritten digits and 700 printed digits.
- Important: both sets contain numerous examples that are ambiguous, unclassifiable, or even misclassified.



Handwritten examples of digits and numbers from the dataset:

40004	75216
14199-2087	23505
96203	14310
44151	<u>05153</u>

Handwritten Digit Recognition

- Handwritten digit recognition with a back-propagation network (Y. Le Cun et al. 1990)

Step 2 – Image pre-processing

- Thresholding, segmentation of individual digits
- Normalisation of digit sizes

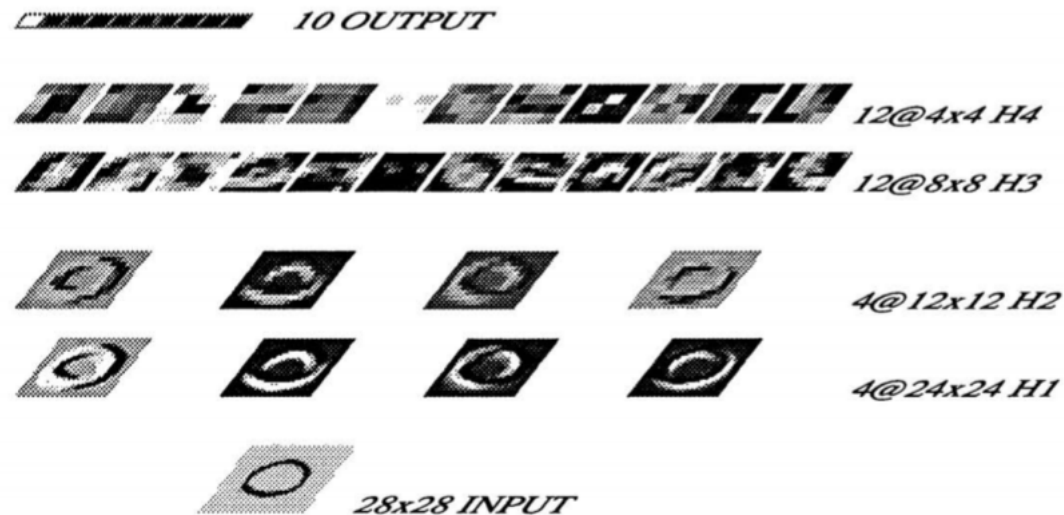


Handwritten Digit Recognition

- Handwritten digit recognition with a back-propagation network (Y. Le Cun et al. 1990)

Step 3 – Create a multi-layer neural network (LeNet-1)

- Input: normalised digit image
- Output: digit category (0-9)



Handwritten Digit Recognition

- Handwritten digit recognition with a back-propagation network (Y. Le Cun et al. 1990)

Step 4 – Apply on the test image

- Pre-processing
- Then apply the trained neuron network to obtain the digit category
- Evaluation using error rate (3.4%) and mean square error (0.024)



Key Point 1

- To design a machine learning system:
 - Need to construct a training set and a testing set
 - Training produces a machine learning / classifier model, which is then applied to the test image to generate the classification output
 - Some image pre-processing might be needed before the training and testing processes

Handwritten Digit Recognition

- Dataset is important
 - Creation of the MNIST database, 1998
 - Training set has 60,000 samples; Test set has 10,000 samples
 - Pre-processed with size normalisation (28x28 pixels) and image centering
 - The large size of this database enables many subsequent studies of more advanced methodologies



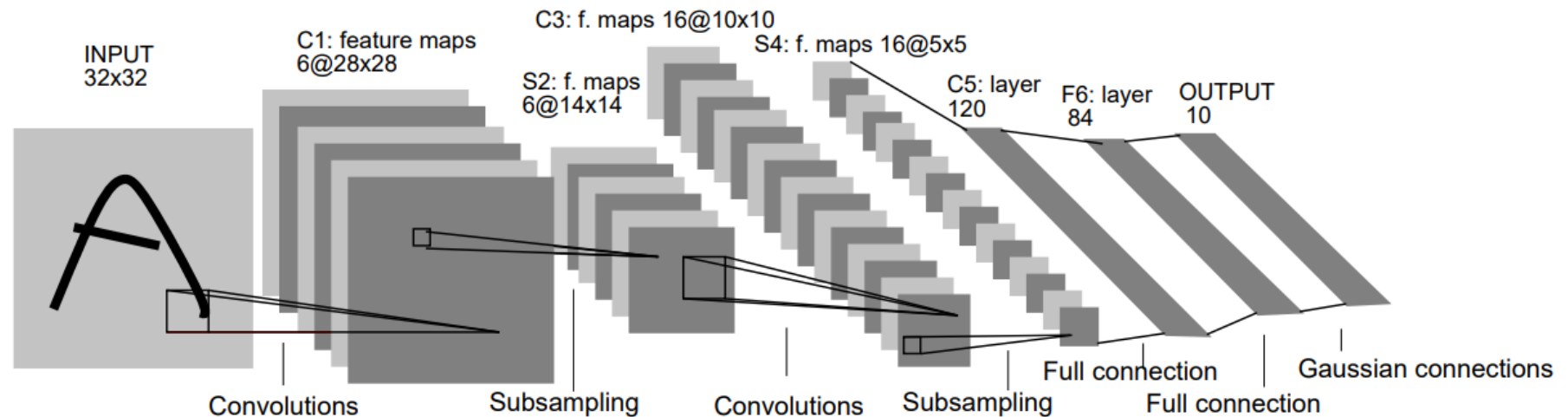
Handwritten Digit Recognition

- Design of machine learning / classifier models is important

Classifier used	Best performance achieved / lowest error rate (%)
Linear classifier	7.6 (LeCun et al. 1998)
Non-linear classifier	3.3 (LeCun et al. 1998)
K-Nearest Neighbour	0.52 (Keysers et al. IEEE TPAMI 2007)
Support Vector Machine	0.56 (DeCoste and Scholkopf, MLJ 2002)
Convolutional Neural Network	0.21 (Wan et al. ICML 2013)

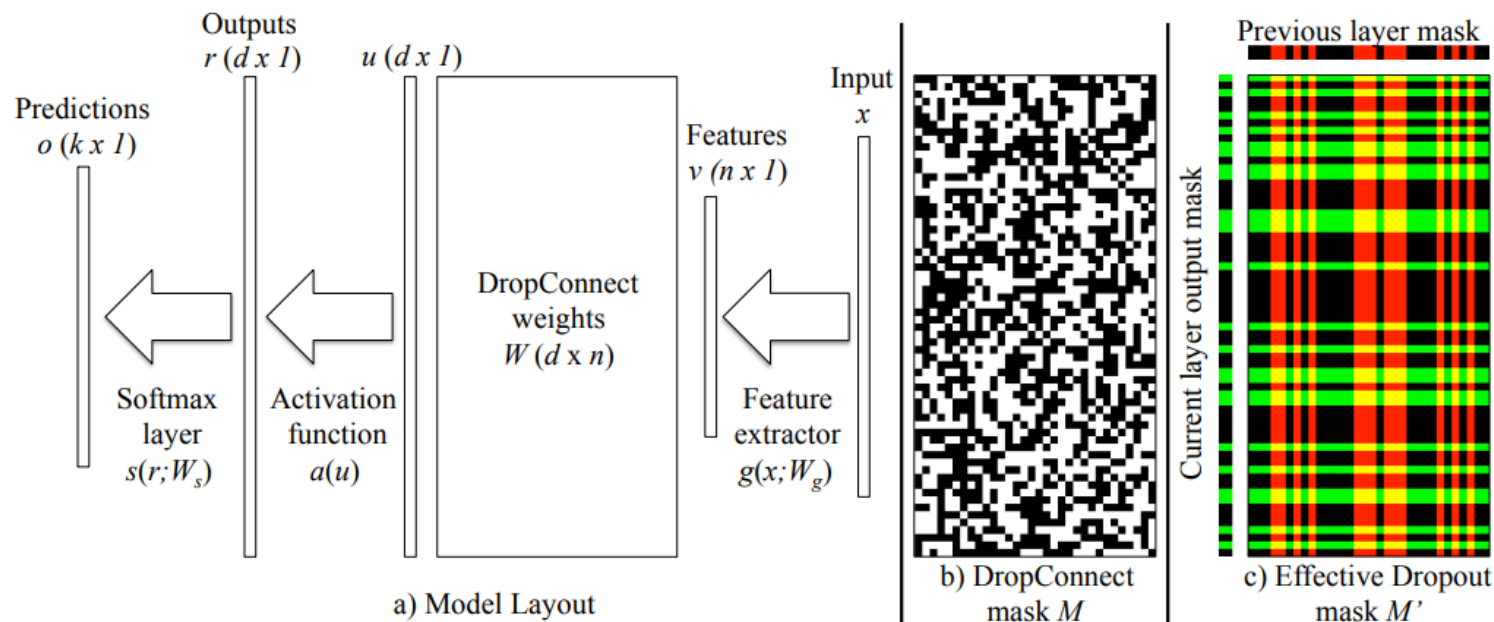
Handwritten Digit Recognition

- Convolutional Neural Network was proposed in 1998
 - Y. LeCun et al. *Gradient-based learning applied to document recognition*. Proc. Of the IEEE, 1998.
 - The CNN network was named “LeNet-5”, giving an error rate of 0.7%.
 - Used to read 10-20% of all bank cheques in US



Handwritten Digit Recognition

- State-of-the-art performance (0.21% error rate) achieved in 2013 by introducing *drop-out* into LeNet-5
 - L. Wan et al. *Regularization of neural networks using DropConnect*. ICML, 2013.



Key Point 2

- To design a machine learning system:
 - Size and quality of the training dataset are critical
 - There are many different choices of classifiers
 - Different classifiers give different performances

Summary

- AI is critical in computer vision applications
- Modern computer vision development often requires machine or deep learning methodologies
- The same principles apply to other application domains:
 - Financial data processing
 - Autonomous driving
 - Mass media
 - Social media
 - Politics
 - Bioinformatics

Summary

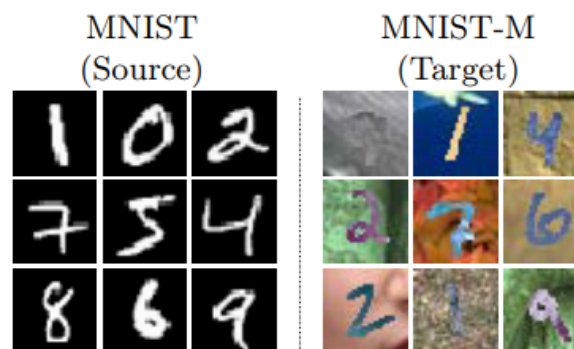
- Major design choices to make:
 - **Pre-processing?**
 - Feature representation?
 - **Classifier?**
 - **Training data?**
 - Evaluation metrics?
- Next:
 - Feature representation
 - Evaluation metrics

Further Reading

- Y. LeCun et al. *Gradient-based learning applied to document recognition*. Proc. Of the IEEE, 1998.
- A brief summary of well-known CNN models
<https://medium.com/@sidereal/cnns-architectures-lenet-alexnet-vgg-googlenet-resnet-and-more-666091488df5>

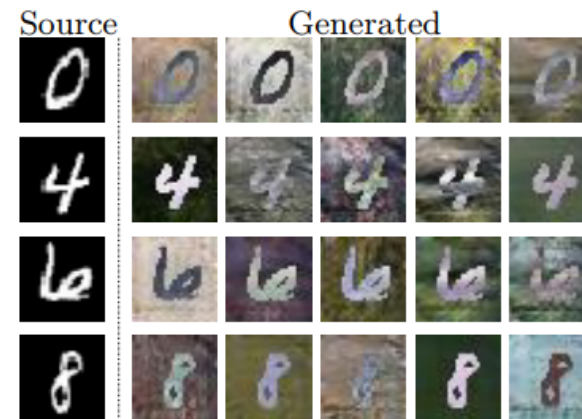
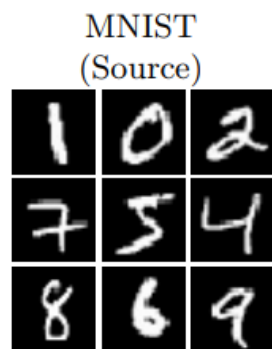
Further Reading

- Go beyond MNIST:
 - MNIST-M dataset was created in 2015 by blending digits from MNIST with colour photos from the BSDS500 dataset
 - The recognition problem becomes much harder
 - The focus of study shifts to *unsupervised domain adaptation*:
 - How to use knowledge learned from MNIST to classify images in MNIST-M?



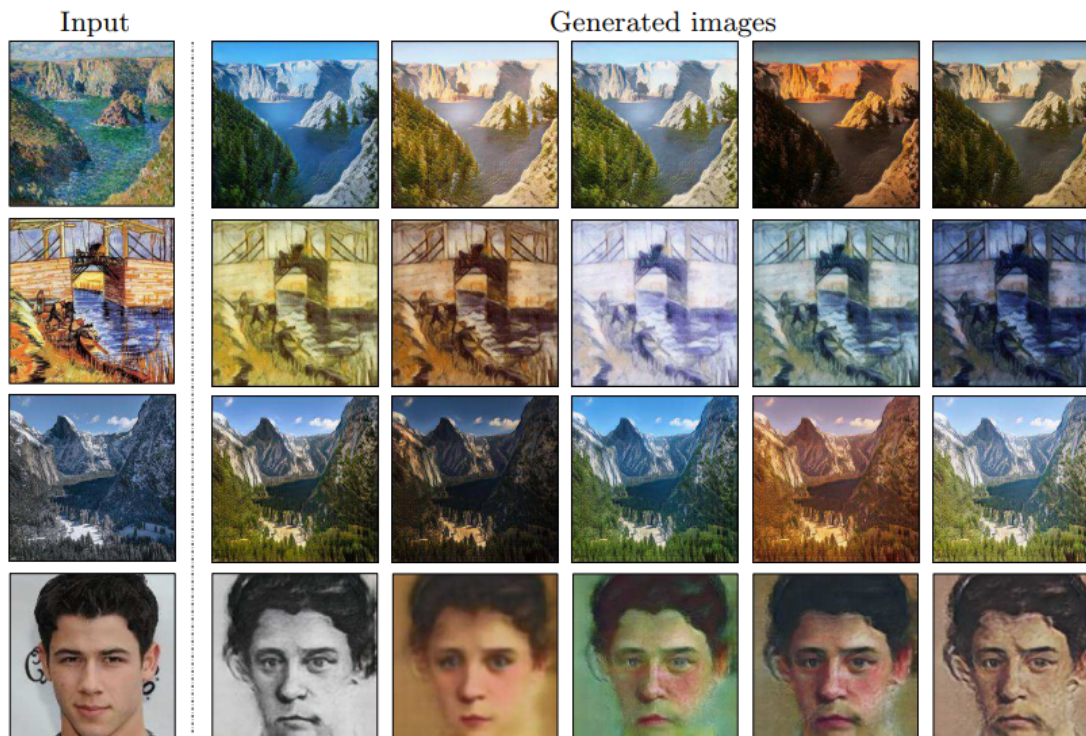
Further Reading

- Unsupervised domain adaptation:
 - H. Lee et al. *Diverse image-to-image translation via disentangled representations*. ECCV, 2018.
 - Training with source images only: 56.6% accuracy
 - Training with target images only: 96.5% accuracy
 - Training on generated target images: 91.54% accuracy



Further Reading

- H. Lee et al. *Diverse image-to-image translation via disentangled representations*. ECCV, 2018.



Monet => Photo

van Gogh => Monet

winter => summer

photo => portrait